claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-acd90fade0d2923cc.jsonl`
- SHA-256: `2c1aa21426e9d11c2cac3c7590817d089939858b81f905950f518287c48e9163`
- Source modified: `2026-06-16T03:27:42+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:26:42.023Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"Service (rally start) is detected as a discrete classification problem using court details and player poses to distinguish the serve from other badminton actions, i.e. rally boundaries are detected by a learned classifier over pose/court features rather than purely from shuttle trajectory."

Source: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15 (primary)
Supporting quote: "It utilizes court details and poses to accurately identify the service action from the other badminton actions."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:26:44.544Z)

I'll verify this claim about the Springer paper. Let me fetch the source to check what it actually says.

3. user (2026-06-16T03:26:51.553Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15"
- prompt: "What is this paper about? Specifically: (1) How does it detect the service/rally start action? Does it use court details and player poses as features in a learned classifier? (2) Is service detection framed as a discrete classification problem? (3) Does it distinguish service from other badminton actions using pose/court features rather than shuttle trajectory? Quote the relevant sentences about service detection methodology, court details, and poses."

4. user (2026-06-16T03:26:57.883Z)

Web search results for query: "badminton service detection court details player poses classification 978-3-031-69073-0_15"

Links: [{"title":"Towards Structured Analysis of Broadcast Badminton Videos","url":"https://arxiv.org/pdf/1712.08714"},{"title":"Automated Service Height Fault Detection Using Computer Vision and Machine Learning for Badminton Matches","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10747833/"},{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"},{"title":"Tracking Players in a Badminton Court by Two Cameras","url":"https://arxiv.org/pdf/2308.04872"},{"title":"Badminton Pose Classification Object Detection Model by BadmintonTutor","url":"https://universe.roboflow.com/badmintontutor/badminton-pose-classification"},{"title":"A Comprehensive Review of Computer Vision in Sports: Open Issues, Future   Trends and Research Directions","url":"https://arxiv.org/pdf/2203.02281"},{"title":"Badminton Detection Using Lightweight Neural Networks for Service Fault Judgement | SpringerLink","url":"https://link.springer.com/chapter/10.1007/978-981-97-2275-4_15"}]

Based on the search results, I found information about the publication you referenced. The work "Service and End of Rally Detection in Badminton Videos" addresses the need to annotate the start and end of rallies in badminton matches.

Video analysis of points won and lost in badminton matches has become a valuable tool for coaches and players to analyze their matches.

The search revealed several related research areas in badminton video analysis:

- Systems using computer vision and machine learning with YOLOv5 object detection can identify elements such as shuttlecocks, rackets, players, and players' shoes

- Player tracking can be achieved through human detectors and trackers combined with perspective transformation, effectively addressing the problem of tracking badminton players with occlusion and overlapping

- Classification systems for tactical movement in badminton have achieved player position detection accuracy of 96.03% and 97.09% on two halves of a badminton court

The ISBN you provided (978-3-031-69073-0_15) corresponds to a chapter in the ISACE 2024 conference proceedings published by Springer.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. assistant (2026-06-16T03:27:00.880Z)

The title is confirmed: "Service and End of Rally Detection in Badminton Videos." Let me get the actual methodology details. Let me try the redirect and search for an open-access version.

6. user (2026-06-16T03:27:04.959Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://idp.springer.com/transit?redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15&code=01dd2648-f97e-4774-82dc-dbc38345e51e
Redirect URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=01dd2648-f97e-4774-82dc-dbc38345e51e
Status: 302 Found

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=01dd2648-f97e-4774-82dc-dbc38345e51e"
- prompt: "Quote the abstract and any methodology text. Specifically: How does it detect the service (rally start) action? Does it use court details and player poses as features in a learned classifier? Is service detection a discrete classification problem? Does it distinguish service from other actions using pose/court features vs shuttle trajectory? Also: how is END of rally detected?"

7. user (2026-06-16T03:27:10.433Z)

Web search results for query: ""Service and End of Rally Detection in Badminton Videos" court poses service action classification method"

Links: [{"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature Link","url":"https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"},{"title":"Automated service scene detection for Badminton game analysis Using CHLAC and MRA | Request PDF","url":"https://www.researchgate.net/publication/290557755_Automated_service_scene_detection_for_Badminton_game_analysis_Using_CHLAC_and_MRA"},{"title":"Detection of service activity in a badminton game | Request PDF","url":"https://www.researchgate.net/publication/261087356_Detection_of_service_activity_in_a_badminton_game"},{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link","url":"https://link.springer.com/article/10.1007/s42979-025-03935-0"},{"title":"Detection of service activity in a badminton game | IEEE Conference Publication | IEEE Xplore","url":"https://ieeexplore.ieee.org/document/6129115/"},{"title":"Ready-to-Serve Detection in Badminton Videos | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Ready-to-Serve-Detection-in-Badminton-Videos-Yue-Paramesran/0744a83d324e9b2d87577ebd5ec5f238744fceb5"},{"title":"Badminton Shot Event Detection and Feature Calculation from 3D Rally Video | Proceedings of the 52nd International Conference on Parallel Processing Workshops","url":"https://dl.acm.org/doi/10.1145/3605731.3605883"},{"title":"Detecting player position during a badminton rally","url":"https://ramanlabs.in/static/tutorial/detecting_player_position_during_a_badminton_rally.html"}]

Based on the search results, I found relevant information about service and end of rally detection in badminton videos. Here's what the research shows:

Overview

A technique has been presented to detect the start of service and end of rally, utilizing court details and poses to accurately identify the service action from other badminton actions.

Methodology

The characteristics of service action are utilized to obtain features and arranged in a 2D format for a convolutional neural network (CNN) to detect service and non-service actions. Additionally, gradients are used to detect the flight of the shuttlecock until it eventually comes to rest which signifies the end of the rally.

Advanced Approaches

More recent research employs sophisticated deep learning methods. For detecting the start of service, two different models are used: Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN), while gradient computation determines the end of the rally. A total of 12 frames with 26 features from each frame comprising distance, angle, gradient and movements typical of serving and receiving actions are arranged in 2D format as inputs to the models.

Performance Results

The experimental results show 88.6% accuracy for a total of 35 rally points.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:27:12.745Z)

The search results give substantial detail about the methodology. Let me confirm the exact details from the paper text and the related comparative paper.

9. user (2026-06-16T03:27:18.500Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/article/10.1007/s42979-025-03935-0
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0"
- prompt: "How does this paper detect start of service and end of rally in badminton? What features (distance, angle, gradient, court, poses)? Which models (BiLSTM, CNN)? Is service detection a classification problem distinguishing service from non-service actions using pose/court features? How is end of rally detected (gradient/shuttle flight)?"

10. user (2026-06-16T03:27:21.153Z)

Service and End of Rally Detection Analysis

Abstract Quote
"It utilizes court details and poses to accurately identify the service action from the other badminton actions."

Detection Methods

Service Detection (Rally Start):
Yes, the approach frames this as a binary classification task. The researchers extract characteristics of service actions and arrange them in 2D format as input to a CNN classifier that distinguishes service from non-service movements. Court geometry and player pose information serve as the foundation for feature extraction.

End of Rally Detection:

This uses a different mechanism?gradient-based shuttlecock flight tracking. The algorithm monitors the movement trajectory of the shuttlecock until it comes to rest, which signals rally conclusion. The method accounts for missing coordinate data and instances when the shuttlecock leaves the camera's view.

Performance

The experimental results showed 88.6% accuracy across 35 rally points in two test videos (4 and 12 minutes duration).

11. assistant (2026-06-16T03:27:32.926Z)

I now have enough to verify the claim. The primary source confirms the claim directly. Let me assess against the checklist.

Checklist assessment:

1. **Supported by quote?** Yes. The abstract states: "It utilizes court details and poses to accurately identify the service action from the other badminton actions." Independent fetches confirm the mechanism: characteristics of the service action are arranged in 2D format as input to a CNN that classifies service vs non-service actions, built on court geometry and player pose features. This is precisely "a learned classifier over pose/court features distinguishing serve from other actions." The claim is well-supported, not an overreach.

2. **Contradicting evidence?** None found. Multiple independent sources (Springer abstract, the related comparative paper SN Computer Science 2025 by apparently the same group) corroborate: service start via CNN/BiLSTM over 26 features per frame (distance, angle, gradient, movements) on 12 frames in 2D format; end of rally via shuttle gradient/flight tracking. Consistent across sources.

3. **Source quality.** Primary peer-reviewed Springer book chapter (ISACE 2024 proceedings), plus a corroborating Springer journal article (SN Computer Science 2025). Source quality matches the claim strength.

4. **Outdated?** No ? 2024 chapter, 2025 follow-on.

5. **Marketing/cherry-picked?** No ? peer-reviewed academic publication.

One minor nuance worth noting (not refuting): the paper detects only the **service/rally-start** boundary via the pose/court classifier; the **end of rally** is detected via shuttle-trajectory gradient, NOT the pose classifier. But the claim is specifically about service/rally-start ("Service (rally start) is detected as a discrete classification problem... rally boundaries"), and the claim's wording ("rally boundaries are detected by a learned classifier over pose/court features rather than purely from shuttle trajectory") is accurate as a contrast for the *start* boundary. The claim does not falsely assert the classifier handles both boundaries. The claim holds.

12. user (2026-06-16T03:27:39.569Z)

Structured output provided successfully

13. assistant (2026-06-16T03:27:41.947Z)

Verification complete. The claim is supported by the primary source and corroborating evidence ? refuted=false, high confidence.